Free Space Optical and Dielectric Characterization for Recycling Sorting

6.2300 Class Project: Detailed Plan
TeamID: RecyclingSorter

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***Abstract*—We present an automated recycling sorting system using a NanoVNA connected to horn antennae to perform a frequency sweep for the recovery of relative permeability and permittivity of a material. Coupled with a stepper motor for mechanical separation this technique allows for sorting metals and plastics.**

I. Introduction & Motivations

Each year, the United States alone generates close to 300 million tons of municipal solid waste (MSW). In 2018, 146.1 million tons of waste were sent to the landfill, and more than 75% of this waste could be processed as recycling or food waste, allowing for material recovery [1]. Our current waste treatment systems are not equipped to properly sort comingled waste streams, leading to large inefficiencies like the one presented above. In addition, high rates of contamination in the recycling stream leads to large amounts of recyclable waste that is not properly processed. Despite many efforts across the United States to improve waste sorting, there has been limited success, and contamination rates are still very high.

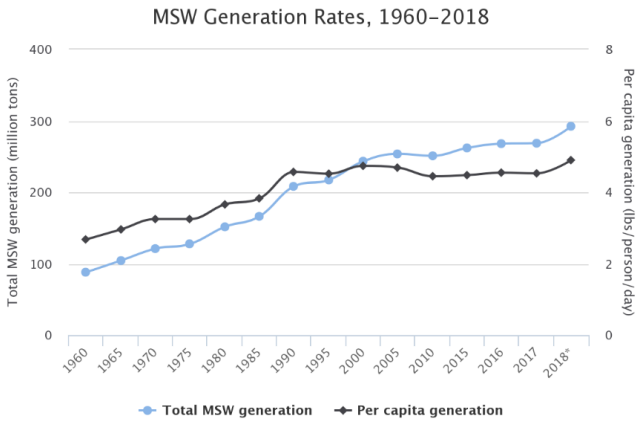

Fig. 1. Municipal Solid Waste (MSW) generation rates have increased since the 1960s (Source: EPA)

One technique that has been explored to address the recycling sorting problem is through imaging of the stream and materials classification through AI image recognition. It has been explored at the research scale and in the waste management industry. One of the most successful AIs has an accuracy rate of 96.96 percent [11].

Another approach that has been used to solve the waste sorting problem is through the use of various sensors including a PIR sensor, LDR sensor, inductive and capacitive proximity sensors, and an ultrasonic sensor all connected to a central microcontroller [10]. This system achieved a 72% accuracy rate for material detection. The process flow includes metal object detection followed by plastic object detection, classifying the material as "neither" if it does not fit the properties of either of the categories.

One relatively unexplored possible waste sorting technique is to compare their dielectric properties. Techniques for characterizing the dielectric properties of materials can generally be divided into two categories: resonant, and non resonant. Resonant methods involve introducing a sample to the inside of a resonator and observing the resulting changes in resonant frequency and quality factor, which can then be used to calculate the permittivity and permeability of the sample. Non-resonant methods derive these same properties of the sample via the changes in impedance and wave velocity that occur when the sample is placed in an obstructing path of some EM radiation - for example, in between two waveguides or a pair of antennae [2]. García, Helena Cano [7] offers a useful experimental comparison between the use of antennae and waveguides for determining the complex permittivity of different glucose concentrations, the results of which suggest that the antenna technique appears to be best adaptable to making measurements quickly and easily outside of a lab setting.

An additional study by Gonçalves et al. [8] provides an informative discussion of different analysis algorithms for obtaining permittivity from the reflection and transmission coefficient data. They compare and contrast the effectiveness of two analytical techniques, multi-frequency extraction and frequency-by-frequency extraction, and conclude that the multi-frequency approach is best for highly noisy data but that the accuracy of frequency-by-frequency extraction can be greatly improved by implementing time-domain gating to isolate the signals of interest in a given exposure cycle.

Considering the strong potential utility indicated in the literature of this free space characterization method for our application of interest, we propose a materials sorting method for recycling classification using reflection and transmission measurements in the UHF frequency band. This process will be carried out using a NanoVNA connected to two horn

antennas as well as an Arduino controlling the stepper motor.

II. Materials & Methods

A. System Overview

The main classification device is a NanoVNA attached to two horn antennas and used to measure the relative permittivity and permeability of a sample placed between the antennae. These components will be connected to a microcontroller which will serve as the control flow unit for this process and send commands to a stepper motor for mechanical separation.

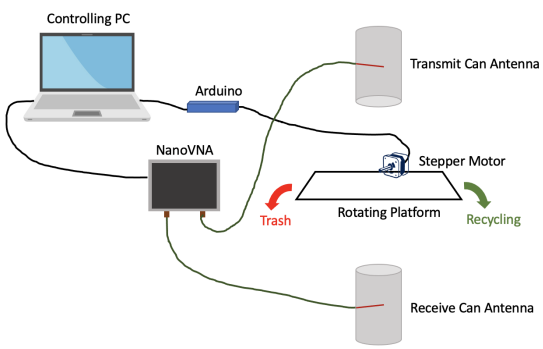

Fig. 2. System schematic detailing setup of waste characterization device.

B. Electrical Characterization

1) Antenna Theory and Design: The permittivity measurement setup consists of two horn antennae, with one connected to the transmit port and the other connected to the receive port of a NanoVNA. We will use the can resonator setup from the previous class lab to construct our antennae, and use the methods covered in class (quarter wave transformer, antenna length trimming, etc) to tune them to minimize power loss by matching operating frequency and impedance.

2) Calibration and Measurement of Scattering Parameters: In addition to the typical SOLT calibration of the NanoVNA, we also follow the free space calibration procedure outlined in Gonçalves et al. [8] and illustrated in Figure 3 to correct for other undesired contributions to the S parameter measurements not from the sample. The scattering parameters $S^{(\text{metal})}$ are obtained by placing a sheet of metal (ideal reflection coefficient=-1) in the sample plane, the scattering parameters $S^{(\text{air})}$ are obtained with the sample plane empty, and finally the scattering parameters $S^{(\text{sample})}$ are obtained by measurement of the sample. The calibrated S_{11} and S_{21} can then be calculated as follows:

$$S_{11} = -\frac{S_{11}^{(\text{sample})} - S_{11}^{(\text{air})}}{S_{11}^{(\text{metal})} - S_{11}^{(\text{air})}} \exp\left(2j(\omega/c)L_1\right) \tag{1}$$

$$S_{21} = \frac{S_{21}^{(\text{sample})} - S_{21}^{(\text{metal})}}{S_{21}^{(\text{air})} - S_{21}^{(\text{metal})}} \exp\left(-j(\omega/c)d\right) \tag{2}$$

where d and L_1 are phase adjusting factors given by the approximate thicknesses of the sample and the metal, respectively.

All of the scattering parameter measurements are done by measuring the magnitude and phase of S_{11} and S_{21} with the NanoVNA in cw sweep mode across our frequency range of 2-3 GHz.

3) Permittivity Calculation: We plan to interface with the NanoVNA via a python script to extract the relative permittivity ϵ_r and relative permeability μ_r from the S parameter data described above. We derive the process for doing this via the theory discussed in class (checked against Gonçalves et al. [8] and García, Helena Cano [7]). For each measurement we plan to perform the following calculations for the values of the S parameters at the frequency corresponding to maximum gain of the horn antenna system. First, we express S_{11} and S_{21} in terms of the reflection coefficient Γ and transmission coefficient T:

$$S_{11} = \frac{(1-T^2)\Gamma}{1-T^2\Gamma^2} \tag{3}$$

$$S_{21} = \frac{(1-\Gamma^2)T}{1-T^2\Gamma^2} \tag{4}$$

We can rearrange these equations to solve for Γ and T:

$$\Gamma = K \pm \sqrt{K^2-1} \tag{5}$$

$$T = \frac{(S_{11}+S_{21}-\Gamma)}{[1-(S_{11}+S_{21})\Gamma]} \tag{6}$$

where $K = \frac{(S_{11}^2 - S_{21}^2 + 1)}{2S_{11}}$ and we take the solution for Γ which satisfies $|\Gamma| \leq 1$. Additionally, we have expressions for Γ and T in terms of the impedance $\sqrt{\frac{\mu_r}{\epsilon_r}}$:

$$\Gamma = \frac{(\sqrt{\mu_r/\epsilon_r}-1)}{(\sqrt{\mu_r/\epsilon_r}+1)} \tag{7}$$

$$T = \exp -j(\omega/c)d\sqrt{mu_r\epsilon_r} \tag{8}$$

Rearranging the above to solve for μ_r and ϵ_r yields the following system of equations:

$$\frac{\mu_r}{\epsilon_r} = [\frac{(1+\Gamma)}{(1-\Gamma)}]^2 = c_1 \tag{9}$$

$$\mu_r\epsilon_r = -[\frac{c}{\omega d}\ln\frac{1}{T}]^2 = c_2 \tag{10}$$

where the two unknowns $\mu_r = \sqrt{c_1c_2}$ and $\epsilon_r = \sqrt{c_2/c_1}$ can be found by substituting in the experimentally determined values of Γ and T extracted from the S parameters using equations 5 and 6.

C. Sorting Mechanism

The sorting mechanism relies on two forms of calibration followed by sample measurement and then sample sorting. The NanoVNA is first calibrated using SOLT calibration and then using the sample plane and sheet metal. It then measures electrical characterization parameters described above to distinguish between plastics and metals. The stepper motor then uses the measured value and established threshold to sort the object as plastic or metal.

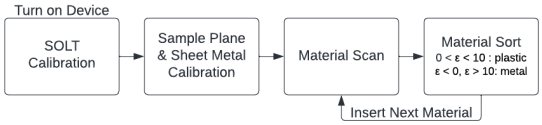

Fig. 3. Process flow used to sort each material.

D. Materials List

- NanoVNA x 1, horn antenna x 2, coaxial cables x 2
- Stepper motor x 1
- Arduino mega x 1
- Device frame, acrylic, screws, wood
- Test materials, see figure 4

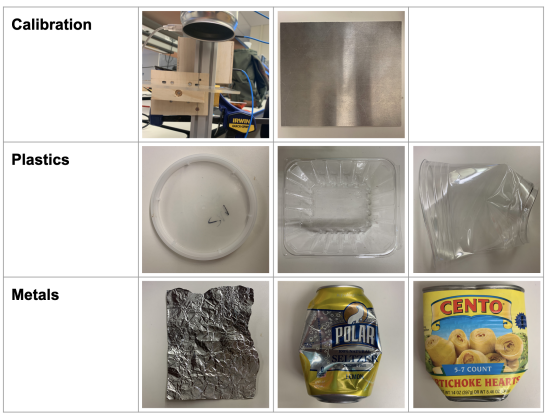

Fig. 4. Materials that were tested to obtain relative permittivity values.

III. Legality, Safety and Ethics

1) Legal Concerns: The FCC limits designers to creation of five transmitters, and this system uses only one transmitter [4]. Our device will operate in the UHF band from 2 to 3 GHz. Some frequency ranges are limited to 200 uV/m. These must adhere to maximum permissible exposure limits for occupational exposure for the RF/microwave wavelengths. From 1.5 GHz to 3 GHz, the power density limit is 5 mW/cm2 [5]. No lasers will be used and no GPS or cell phone jamming will be performed [3][6].

2) Safety Concerns: Potential safety issues with our system include the stepper motor heating up during operation to temperatures above 80°C which could cause minor burns [9], limbs or other body parts being scanned by low power electromagnetic waves, or the rotating platform hitting the user. We have designed the system to encapsulate the motor and have all electrical components away from the area that the user would access to use the system. The stepper motor turns slowly and the sample platform is small so it will not hit the user hard if it does. The device will not rest directly on the waste bins so as to minimize the dangers posed to staff members when removing the bins from below to empty them.

3) Ethical Concerns: Several stakeholders could be negatively impacted by this project. If waste sorting shifts towards more source separation, municipal-scale waste sorting infrastructure will no longer be required. The industry could adopt new technologies such as our proposed system. Those who work in waste sorting education will have less demand for their expertise.

At scale, this system would strengthen the market for recycled plastic by increasing supply. Competing markets - for example, the fossil fuel/refinery industry - would be negatively impacted because more recycled plastic means less virgin plastic is needed. Most reasonable actors would agree, however, that this is a positive externality.

IV. Results

Plastics and metals exhibited different S parameter values as well as real parts of permittivities. As can be seen in figure 5, the S21 (red) magnitude for plastics is relatively stable and positive throughout the whole frequency range. In addition, the real part of the permittivity is constant except for spikes in the 2 to 2.15 GHz and 2.5 to 2.7 GHz ranges. In metals, however, as seen in figure 6, both the S11 (blue) and S21 (red) parameters fluctuate over the frequency range, and the S21 parameter is mostly negative. The real part of the permittivity exhibits an increase starting at higher frequencies.

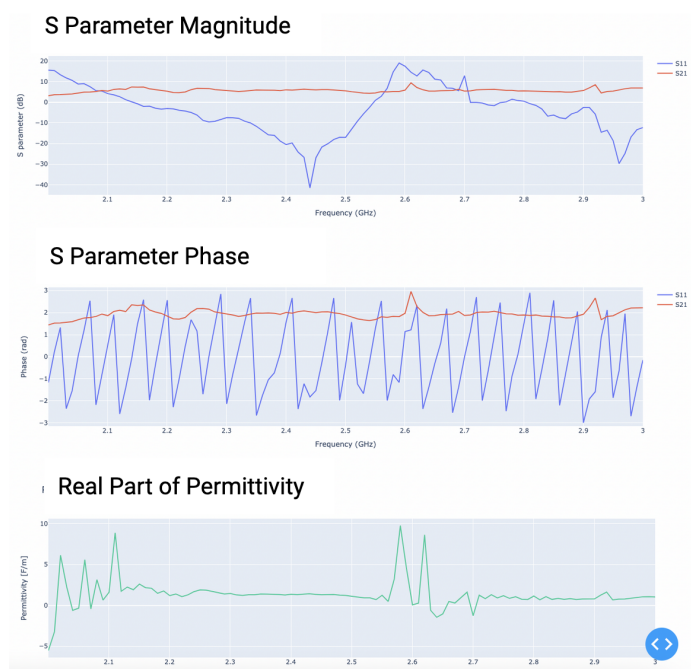

Fig. 5. Curves of measured S11 (blue) and S21 (red) magnitudes and phases for a plastic material. Calculated permittivity curve based on the S11 and S21 magnitudes.

In order for this method to be applied at an industry scale, accuracy rate is a crucial metric. We evaluated two calibration objects, three plastics, and three metals to determine if they were properly sorted basted on the threshold bounds.

Thresholds were established at relative permittivity values of 0 and 10. Values below zero and above 10 fall into the metals category, while values within the range fall into the plastics

Fig. 6. Curves of measured S11 (blue) and S21 (red) magnitudes and phases for a metal material. Calculated permittivity curve based on the S11 and S21 magnitudes.

TABLE I
MATERIAL PERMITTIVITY VALUES

Category	Item	n	Permittivity
Calibration	Acrylic Plane	15	1.27 ± 0.00561
Calibration	Sheet Metal	15	14.1 ± 0.000894
Plastics	LDPE Lid	15	1.34 ± 0.00368
Plastics	PETE Container	15	1.29 ± 0.00304
Plastics	PS Cup	15	1.32 ± 0.00129
Metals	Aluminum Foil	15	66.6 ± 0.000231
Metals	Aluminum Can	15	544 ± 0.000275
Metals	Steel Can	15	331 ± 0.00575

category. All of the items that we tested are accurately sorted using these threshold bounds.

These results are also demonstrated in Figure 7, which shows the permittivity values for (a) Linear scale and (b) Logarithmic scale with the plastic threshold region highlighted in blue. For these tests, the plastics and metals fall within the correct bounds. Scans of each item were performed 15 times and the mean and standard deviations were calculated, as shown in Table I.

V. DISCUSSION

This system calculated relative permittivity values for plastics ranging from 1.25-1.35. Literature values demonstrate values of 2.3 for LDPE, 3-4 for PS, and 2.4-2.7 for PETE [12]. For metals, permittivity values above 50 were calculated by the system. The theoretical permittivity of metals is infinite, and the calculated values are approaching this bound. Although

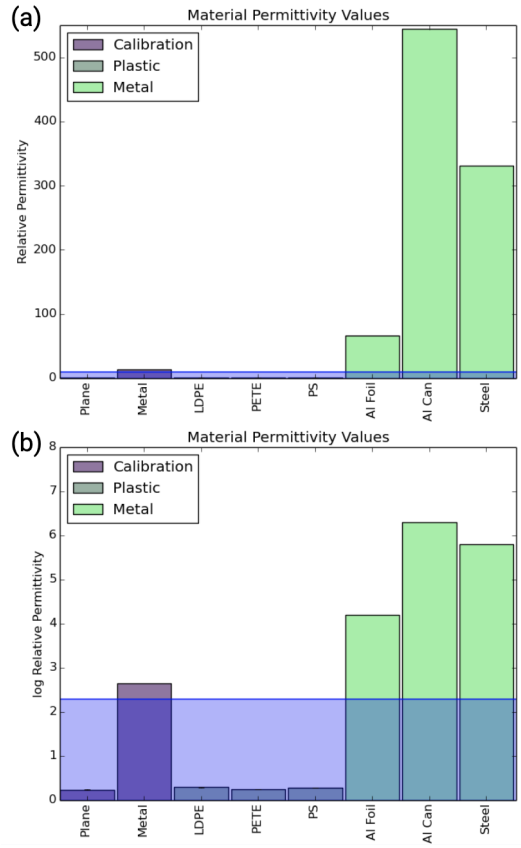

Fig. 7. Relative permittivity values of tested materials in (a) Linear scale and (b) Logarithmic scale. Error bars are included but extremely small. The blue region indicates where the materials are sorted as plastics.

calculated values are slightly lower than in the literature, the system demonstrates accuracy in distinguishing between plastics and metals.

In order to increase system applicability, expanding the scope of sorting to distinguishing between different types of plastics and other materials is important. This could be done through the creation of a system that relies on the same principles but uses higher quality equipment to obtain permittivity values with very low errors. Including sample dimensions and geometries into measurements willl also increase accuracy. Once extensive testing is done, highly accurate thresholds for different materials types could be established. In addition, adding in a machine learning feature which allows for training using different materials could expand the application scope.

VI. CONCLUSIONS

Using horn antennae to analyze the dielectric properties of materials, we were able to accurately distinguish between metals and plastics to obtain clearly different values for relative permittivities, with values for plastics ranging from 1.25 to 1.35 and for metals from 66 to 544. A relative permittivity threshold of 10 accurately sorts between these different material types using an automated stepper motor system in a repeatable fashion with low errors.

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